🔐 XencyPass Password Manager

Complete Project Documentation & System Architecture | Secure Password & Document Manager for Android Version 1.0 | Last Updated: March 2026

📋 1. Project Overview

XencyPass is a secure password manager and document wallet for Android devices. It allows users to store login credentials, important documents, and other sensitive information with military-grade encryption.

🎯 **Mission** To provide a secure, user-friendly platform where users can store and manage their digital credentials safely, with built-in recovery options for lost access.

- 🔐 **Zero-Knowledge Architecture:** All encryption happens on-device. Even we cannot access your passwords.
- 📸 **Smart Document Scanner:** Upload documents with auto-crop feature for IDs, passports, and cards.
- 👥 **Trusted Recovery System:** Recover your account safely through manual verification and trusted contacts.
- ☁️ **Dual Backup System:** Google Drive for automated full backups + Firebase for exclusive emergency access.

✨ 2. Key Features

🔐 Authentication & Security (Adaptive Login)

- **Master Password Only Login** - Primary authentication method (No email required for daily access)
- **Adaptive Risk Assessment** - Real-time evaluation of Device Fingerprint and Risk Score
- **Step-Up Biometric Verification** - Fingerprint scan required if risk score fluctuates
- **Email OTP Fallback** - Verification code sent if hand biometrics are not set up
- **Smart Lockout System** - Account safely locks for 30 minutes after 3 failed Master Password attempts
- **Auto-Lock Timer** - Configurable timeout (1 min to 30 mins)

🔒 Password Management

- **Store Login Credentials (App OR Website)** - Categorized securely. Fields include: App or Platform Name, Email/Username, Password, 8-Digit Code, Secret Phrase, Link, and Note.
- **Smart Auto-Fill & Categorization** - User selects if the credential is for an App or Website.
  - If **App**: The Link field is hidden, and the title becomes "App Name".
  - If **Website**: The Link field is visible ("Website Name").
  - As the user types the Platform Name, our system automatically categorizes it and auto-fills the real website URL. If the system cannot find the URL automatically, it

shows a short error prompting the user to "Please add Website URL" manually.

- **Secure Password Generator** - A simple 1-click "Generate" button placed on the right side of the password field to automatically generate and fill a secure password.
- **Favorites & Important Marks** -
  - **Favorites**: Pins items to the top of the list for regular, easy access.
  - **Important Mark**: Triggers the system to automatically store a secure backup of this specific credential into the Firebase Emergency Vault.
- **Clipboard Auto-Clear** - Passwords clear from clipboard after 30 seconds.

📄 **Document Wallet**

- **Multiple Document Types** - National ID, Passport, Driver's License, Credit Cards, and **Online Copies** (to support modern e-documents globally).
- **Smart Camera Scanner** - The user snaps a picture of their physical document. The system automatically marks the 4 edges, fixes the tilt, and straightens the image. It provides an edit option so the user can manually adjust the tilt and crop if there are any minor fluctuations.
- **Secure Image Storage** - Encrypted image storage.

☁️ **Backup & Sync**

- **Tiered Google Drive Backup** - Automated backups based on user plans:
  - **Free Users**: 1 Backup per day.
  - **Pro Users**: 10 Backups per day.
  - **Advance Users**: 25 Backups per day.
- **Firebase Emergency Vault** - Only stores items that the user has explicitly marked as **"Important"**.
- **Zero-Corruption Compression** - Highly optimized compression algorithm ensures the data size is reduced while maintaining a strict 0% data corruption rate.

👥 **Trusted Recovery System (DMS)**

- **Inactivity Detection** - Triggered when the user does not log in for a specific period:
  - **Free Users**: 180 Days limit.
  - **Pro Users**: Can choose between 30 to 90 Days.
  - **Advance Users**: Custom time limit (Minimum 2 weeks).
- **Trusted Contact Setup** -
  - **Free Users**: Can add a Friend or Family member via standard email/phone.
  - **Pro/Advance Users**: Can set up a **WhatsApp Trusted Contact** via WhatsApp API (can be their own alternate WhatsApp number or someone else's).
- **Security Questions** - Configured during initial User Sign-Up for future verification.
- **Manual Verification Team** - In case of recovery, our recovery team manually verifies the user request (Response time: 24 hours to 7 days).

- **Re-Login Access Option** - Designed specifically for users who forget their Master Password but still have access to their registered email. After successful manual verification, the user gets a secure Re-Login link to safely access and reset their vault.

💰 Monetization

- **Google AdMob Integration** - Non-intrusive ads for free users.
- **Premium Subscription** - Pro & Advance tiers for additional features, WhatsApp API access, and increased backups.

🛠️ 3. Technology Stack & Dependencies

Tech Stack: Kotlin | Latest Android Studio | Firebase Auth | Firebase Firestore | Firebase Storage | Room Database | SQLCipher | AES-256-GCM | BiometricPrompt API | Google Drive API | WhatsApp API | AdMob | uCrop (Image Cropping) | Glide | Coroutines

📦 Dependencies (build.gradle.kts)

```
// Core Android
implementation("androidx.core:core-ktx:1.12.0")
implementation("androidx.appcompat:appcompat:1.6.1")
implementation("com.google.android.material:material:1.11.0")

// Firebase
implementation(platform("com.google.firebase:firebase-bom:32.7.4"))
implementation("com.google.firebase:firebase-auth-ktx")
implementation("com.google.firebase:firebase-firestore-ktx")
implementation("com.google.firebase:firebase-storage-ktx")

// Biometric
implementation("androidx.biometric:biometric:1.1.0")

// Encryption - SQLCipher
implementation("net.zetetic:android-database-sqlcipher:4.5.4")

// Room Database
implementation("androidx.room:room-runtime:2.6.1")
implementation("androidx.room:room-ktx:2.6.1")
ksp("androidx.room:room-compiler:2.6.1")

// Image Cropping & Edge Detection
implementation("com.github.yalantis:ucrop:2.2.8")

// Google Ads
implementation("com.google.android.gms:play-services-ads:22.6.0")
```

🛡️ 4. 7-Layer Security Architecture

- **Layer 7: Biometric Authentication:** Hardware-backed security | TEE (Trusted Execution Environment)
- **Layer 6: Master Password:** Argon2id hash | Memory-hard 64MB | OWASP recommended
- **Layer 5: Device Binding:** Unique Device ID | Android Keystore System
- **Layer 4: Data Encryption:** AES-256-GCM | Per-field encryption | Random IV per record

- **Layer 3: Database Encryption:** SQLCipher | Full database encryption | 256-bit AES
- **Layer 2: Cloud Double Encryption:** App-level Master Key | Zero-knowledge architecture
- **Layer 1: Transport Security:** TLS 1.3 | Certificate pinning | HSTS enabled

🔐 Encryption Flow

```
User Input (Plain Text)
        ↓
AES-256-GCM Encryption (Master Password Key)
        ↓
SQLCipher Database Encryption
        ↓
Local Storage (Encrypted)
        ↓
If Important: 0% Corruption Compression + Double Encryption
        ↓
Firebase Emergency Vault
```

📂 5. Complete File Structure

```
XencyPassCredentialsManager/
├── app/
│   ├── src/
│   │   ├── main/
│   │   │   ├── kotlin/com/xencypro/xencypass/
│   │   │   │   ├── XencyPassApplication.kt
│   │   │   │   ├── MainActivity.kt
│   │   │   │   ├── auth/
│   │   │   │   │   ├── LoginActivity.kt
│   │   │   │   │   ├── RegisterActivity.kt
│   │   │   │   │   └── BiometricHelper.kt
│   │   │   │   ├── vault/
│   │   │   │   │   ├── VaultActivity.kt
│   │   │   │   │   ├── AddPasswordActivity.kt
│   │   │   │   │   ├── AddDocumentActivity.kt
│   │   │   │   │   ├── SmartScannerHelper.kt
│   │   │   │   │   └── PlatformUrlMatcher.kt
│   │   │   │   ├── recovery/
│   │   │   │   │   ├── RecoverySetupActivity.kt
│   │   │   │   │   └── ManualVerificationActivity.kt
│   │   │   │   ├── encryption/
│   │   │   │   │   └── EncryptionManager.kt
│   │   │   │   ├── database/
│   │   │   │   │   ├── AppDatabase.kt
│   │   │   │   │   ├── PasswordDao.kt
│   │   │   │   │   └── DocumentDao.kt
│   │   │   │   ├── firebase/
│   │   │   │   │   └── EmergencyVaultManager.kt
│   │   │   │   ├── ads/
│   │   │   │   │   └── AdManager.kt
│   │   │   │   ├── models/
│   │   │   │   │   ├── Password.kt
│   │   │   │   │   └── Document.kt
│   │   │   │   └── utils/
│   │   │   │       └── PreferencesManager.kt
│   │   │   ├── res/
│   │   │   │   ├── layout/
│   │   │   │   │   ├── activity_login.xml
```

```
│   │   │   │   │   ├── activity_register.xml
│   │   │   │   │   ├── activity_vault.xml
│   │   │   │   │   ├── activity_add_password.xml
│   │   │   │   │   └── activity_add_document.xml
│   │   │   │   ├── values/
│   │   │   │   │   ├── colors.xml
│   │   │   │   │   ├── strings.xml
│   │   │   │   │   ├── themes.xml
│   │   │   │   │   └── dimens.xml
│   │   │   │   └── drawable/
│   │   │   └── AndroidManifest.xml
│   │   └── google-services.json
│   └── build.gradle.kts
├── build.gradle.kts
├── settings.gradle.kts
└── local.properties
```

🚀 6. Complete Setup Guide

Step 1: Prerequisites

- Latest Android Studio
- JDK 17 or later
- Android SDK API 34
- Google account for Firebase
- Android device or emulator (Android 7.0+ / API 24+)

Step 2: Firebase Setup

1. Go to Firebase Console
2. Create new project → "XencyPass"
3. Add Android app with package name: `com.xencypro.xencypass`
4. Download `google-services.json` and place in `app/` folder
5. Enable Authentication → Email/Password
6. Enable Firestore Database (start in test mode, then add security rules)
7. Enable Storage (for document images)

Step 3: Firestore Security Rules

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    match /users/{userId}/{document=**} {
      allow read, write: if request.auth != null && request.auth.uid == userId;
    }

    match /emergency_vault/{userId}/{itemId} {
      allow read, write: if request.auth != null && request.auth.uid == userId;
    }

    match /recovery_requests/{requestId} {
      allow read, write: if request.auth != null &&
        (request.auth.uid == resource.data.userId ||
```

```
            request.auth.token.admin == true);
        }
    }
}
```

Step 4: Android Studio Setup

1. Open Android Studio
2. Open existing project → Select XencyPassCredentialsManager folder
3. Wait for Gradle sync to complete
4. If sync fails, check `build.gradle.kts` files

Step 5: Run the App

1. Connect Android device (USB debugging enabled) or start emulator
2. Click Run (▶️) or press Shift + F10
3. Select device
4. Wait for build and install

📅 7. Development Roadmap

- **Phase 1: Project Setup & UI/UX Design**
  - Android Studio & Gradle Setup.
  - Basic Screens layout (Splash Screen, Onboarding, Authentication Screens).
  - Design logic for dark/light themes and intuitive UI flow.
- **Phase 2: Authentication & Security Core**
  - Master Password Logic implementation.
  - Device Fingerprint & Risk Score API integrations.
  - Step-up Biometric Authentication (Fingerprint fallback).
  - Lockout logic (30-min block on 3 wrong attempts).
  - Setup Security Questions during Registration.
- **Phase 3: Core Database & Vault API Setup**
  - SQLCipher integration for local encryption.
  - Room Database schema setup (Passwords, Documents).
  - AES-256-GCM encryption logic for all user inputs.
- **Phase 4: Password Management Features**
  - App vs Website form logic (conditional UI rendering).
  - Smart Platform matching API (Auto-fill URLs based on names).
  - 1-Click Secure Password Generator logic.
  - Category sorting, Search, Favorites (Pin to top), and Important tags.
- **Phase 5: Document Scanner Integration**

    - Camera API setup for physical documents.
    - Auto Edge Detection & Deskewing algorithm.
    - Manual crop/tilt adjustment UI for users.
    - Online Copy (e-docs) import logic.
- **Phase 6: Advanced Backup Systems**
    - Google Drive API Integration with Tier restrictions (1/10/25 per day).
    - Firebase Emergency Vault implementation (Only "Important" marked items).
    - Zero-Corruption Data Compression logic before uploading.
- **Phase 7: Trusted Recovery & Monetization**
    - Manual Recovery Request system (24h-7d review process).
    - Inactivity detector implementation (Free/Pro/Adv limits).
    - WhatsApp API Integration for Pro/Advance Trusted Contacts.
    - AdMob integration for Free tier.
    - Billing setup for Pro and Advance premium users.
- **Phase 8: Final Testing & Release**
    - QA Testing on multiple devices.
    - Penetration testing for Master Key encryption.
    - Play Store deployment & Marketing rollout.

📊 8. Complete Data Flow Summary

Data Flow Pipeline: 📝 User Input (Plain text data) → 🔐 Encryption (AES-256-GCM) → 💾 Local DB (SQLCipher)

⬇️

❓ **Important Flag Check:** Is this item marked as important?

- ⭐ **YES:** 📦 Compress (Zero-Corruption) ⬇️ 🔐🔐 Double Encrypt (App Master Key) ⬇️ ☁️ Firebase (Emergency Vault)
- 📦 **NO:** 📤 Backup Queue (Pending sync) ⬇️ ⏰ WorkManager (Scheduled Google Drive Task based on user tier) ⬇️ 📦 Compress (Zero-Corruption)

⬇️

☁️ Google Drive (appDataFolder) → 📊 Firestore (Metadata only)

📈 Data Flow Statistics

- 📊 **Compression Rates:** 0% Data Corruption Rate with highly optimized GZip Level 9 algorithm. Average entry size reduced drastically ensuring high speed and robust security.
- ⚡ **Performance Metrics:** Encryption Speed: Per item: <10ms, Batch 100 items: <500ms, Full database: <2 seconds. Backup Upload: WiFi: ~30 seconds (1MB), 4G: ~60 seconds (1MB). Recovery Download: WiFi: ~20 seconds, 4G: ~40 seconds, Decryption: ~5 seconds.

🔄 **Recovery Flow** 📥 Download (From Drive / Vault) → 🔒 Decrypt (Master key) → 📦 Decompress (Zero-Corruption Extract) → 💾 Restore (Local DB)

⚙️ 9. Detailed System Flows

🔐 User Authentication, Onboarding & Setup Flow

1. 📱 **App Launch:** Splash Screen displays for 2 seconds. XencyPass Logo with smooth animation. Checking authentication status...
2. 📝 **Sign Up Process:** User enters basic account details (Email).
3. ✉️ **Code Verification:** System sends a verification code to the registered email. User must enter this code to verify their identity.
4. 🔑 **Master Password Setup:** After successful email verification, the required Master Password setup screen appears.
   - A short description is displayed explaining why this key is crucial.
   - Includes a wrapped text with a highlighted link: <mark>**Learn More**</mark> (clicking this opens detailed info about zero-knowledge architecture).
   - **Requirements:** Minimum 12 characters, Uppercase + Lowercase letters, Numbers (0-9), Special characters (!@#$%).
5. ❓ **Security Questions Setup:** User is prompted to select and answer security questions. These will be used later if manual verification is ever required.
6. 👆 **Setup Biometric (Optional but highly recommended):** Adding device fingerprint. Security Class: BIOMETRIC_STRONG (Class 3).
7. 💾 **Create Encrypted Database:** Technology: SQLCipher with AES-256, Room Database framework, Auto-migration support.

🚪 Adaptive Smart Login Flow

1. 🔑 **Login Screen:** User navigates to the login screen and enters ONLY their **Master Password**.
2. 🛡️ **System Verification Check:** Behind the scenes, the system checks the user's **Device Fingerprint** and **Risk Score** (similar to Google/Facebook).
3. 🟢 **Scenario A (All Clear):** If both Device Fingerprint and Risk Score match expected normal parameters → User is logged in successfully.
4. 🟡 **Scenario B (Risk Detected):** If the Device Fingerprint is correct but there is a fluctuation in the Risk Score (e.g., unusual time, new IP):
   - The system prompts for **Mobile Fingerprint Verification**.
   - *Reasoning:* Every person has their own unique hand fingerprint identity.
5. 🟠 **Scenario C (Fallback Verification):** If the user does not have a fingerprint (biometric) set up on their device during a Risk Event:
   - The system sends an **Email Verification Code** to the user.
   - User must enter the code to complete login.

6. 🔴 **Maximum Wrong Attempts & Lockout:**
    - A user can enter the wrong Master Password a maximum of **3 times**.
    - If the 3rd attempt is wrong, the account is **locked for 30 minutes**.
    - After 30 minutes, they can try again.

🔒 Password & Document Management Flow

1. 🏠 **Home Screen:** Available Tabs: 📄 Passwords | Wallet | Notes. Features: 🔍 Search bar (instant filter), ☰ Menu (settings, backup), ➕ FAB (add new item).
2. 📋 **Choose Type:** Item Types: 🔑 App/Website Password, 💳 Credit/Debit Card, 🆔 Physical ID Document, 📄 Online E-Document, 🏦 Bank Account.
3. 📝 **Fill Password Form:** * Choose: **App** or **Website**.
    - If Website: Link field shows. Enter Platform Name -> System auto-fills real link. (Shows error if manual entry needed).
    - Fill Username/Email, 8 Digit Code, Secret Phrase, Note.
    - Use the **1-Click Generate Password** button on the right of the password field.
    - Mark as **Favorite** (Pins to top) or **Important** (Saves to Firebase Vault).
4. 📸 **Fill Document Form:**
    - Snap physical document via Camera.
    - System detects 4 edges and auto-corrects tilt/straightness.
    - User manually adjusts crop if needed.
5. 🔐 **Encryption Process:** Algorithm: AES-256-GCM. Process: Generate random IV (12 bytes), Encrypt plaintext data, Generate auth tag (16 bytes), Store encrypted blob. Key Derivation: Master Password → Argon2id → PBKDF2 → 256-bit encryption key.
6. 💾 **Save to Database:** Storage Layer: Room Database (SQLCipher). Tables: passwords, documents, notes, categories, tags. Encryption: Double-layer protection.

☁️ Backup System Flow

1. ⏰ **Backup Triggers:** Automatic Backup based on User Tier limits (Free: 1/day, Pro: 10/day, Adv: 25/day). Manual Backup (User taps "Backup Now").
2. 🔍 **Pre-Backup Check:** Conditions Verified: Network, Battery (>20%), Storage, Google account connected, Backup limit status.
3. 📤 **Export Database:** Exported Tables: passwords, documents, notes, categories, settings, metadata. Format: JSON structure.
4. 🔐 **Encrypt Backup:** Algorithm: AES-256-GCM. Key Derivation: User Master Password + Device Salt → PBKDF2 (10,000 rounds) → 256-bit key. Output: encrypted.bin.
5. 📦 **Compress Data:** Compression: Zero-Corruption algorithm minimizing size securely.
6. ☁️ **Upload to Google Drive:** Location: appDataFolder (hidden from user). API: Google Drive REST API v3.

7. 📊 **Store Metadata:** Firestore Document: users/{uid}/backups/{backupId}. Purpose: Backup history, Version tracking, Recovery validation.

📥 Manual Recovery & Trusted Contact Flow

1. ⏰ **Inactivity Detection:** Trigger Condition based on user tier (Free: 180 days, Pro: 30-90 days, Adv: 2w+).
2. 📧 **Trusted Contact Notification:** * Free Users: Email sent to standard Trusted Contact (Friends/Family).
   - Pro/Adv Users: Message sent securely via WhatsApp API to the registered WhatsApp trusted contact.
3. 🕵️ **User Requested Recovery (Forgot Master Password):** User initiates a recovery request from the app utilizing their registered email.
4. 🛡️ **Manual Verification Team:** The request enters the manual review queue. The recovery team reviews security question answers and device metrics.
   - **SLA:** 24 Hours to 7 Days.
5. 🎉 **Recovery Success & Re-Login:** If verification is successful, the user is provided a secure **Re-Login Option**. This allows the user to regain entry to their vault safely to reset their master password without losing their previously saved credentials.

🛡️ Extended Security Details

- 📝 **User Input Data:** Plain text data in memory must be encrypted before storage to prevent memory dump attacks.
- 🔑 **Master Password Derivation:** Algorithm: Argon2id (Inputs: User master password, Random salt (32 bytes), Iterations: 3, Memory: 64 MB, Parallelism: 4 threads). Intentionally slow!
- 🔐 **Generate Encryption Key:** Algorithm: PBKDF2-HMAC-SHA256 (Inputs: Argon2id derived key, Device unique ID, Salt (32 bytes), Iterations: 10,000). Output: 256-bit AES encryption key. Storage: Android Keystore System.
- 🔒 **AES-256-GCM Encryption:** Galois/Counter Mode Benefits: Authenticated encryption, Parallel processing, NIST approved.
- 💾 **SQLCipher Database:** Database File: xencypass.db (encrypted). Cannot be opened without Master password, Device ID, SQLCipher passphrase.
- 🔐🔐 **Cloud Double Encryption:** Key Source: Firebase Remote Config (Rotated every 90 days). Process: Encrypted data → Encrypt again with app key → Double-encrypted blob → Upload to cloud. Decryption Requires: App master key (cloud), User master password, Device binding key.
- ☁️ **Cloud Storage:** Transport Security: TLS 1.3 encryption, Certificate pinning, Perfect forward secrecy. Access Control: Firebase Auth required, API key obfuscated, ProGuard R8 enabled, Root detection, SSL pinning.
- 🔑 **Key Retrieval:** Source: Android Keystore System. Key Alias: "xencypass_master_key".

- 🔒 **Hardware Decryption:** TEE Process: Retrieve Cipher from Keystore → Initialize: DECRYPT_MODE → Decrypt in secure hardware → Return plaintext to app. Protection Against: Memory dumps, Debugger access, Root exploits, OS vulnerabilities.
- ✅ **Vault Unlocked:** Decrypt SQLCipher key → Open database connection → Decrypt AES-GCM fields → Load data into memory. Security Measures: Clipboard auto-clear (30s), Screenshot prevention, Screen recording blocked, Task manager blur.
- 🚫 **Failed Authentication:** Account Locked safely for 30 minutes after 3 Failed Master Password Attempts! Protects against brute-force guessing attacks.

⚙️ Settings & Configuration

- 🔐 **Security Settings:** Auto-Lock Timer, Biometric Unlock, Master Password, Manual Recovery Status, Security Audit Log.
- ☁️ **Backup Settings:** Automatic Backup, Manual Backup, Google Drive Account, Backup History, Firebase Important Vault.
- 🎨 **Display Settings:** Theme (Light/Dark/Auto/AMOLED), Language, Font Size, Accent Color, Compact View, Password Display (Hidden/Visible).
- ℹ️ **About XencyPass:** App Version, Developer, Privacy Policy, Terms of Service, Open Source Licenses, Contact Support, Rate Us, App Statistics.

📌 10. Important Notes

🚨 Security Reminders:

- Never store plaintext passwords anywhere
- All encryption keys must be hardware-backed (Android Keystore)
- Never log sensitive data
- Implement SSL pinning for network calls
- Regular security audits required

💰 Firebase Cost Considerations:

- Firebase free tier: 50K reads/day, 20K writes/day, 1GB storage
- Suitable for 1-2k daily active users
- Blaze plan needed beyond free tier
- Optimize Firestore reads with caching

📱 Android Compatibility:

- Min SDK: 24 (Android 7.0 Nougat)
- Target SDK: 34 (Android 14)
- Biometric requires Android 6.0+ (API 23)
- Camera features require API 21+